

Grounding Without Corrective Control: Truth-Tracking Profiles for Large Language Models

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Abstract

Recent work suggests that some large language model representations have content or reference. A different question remains. Grounding can secure either without supplying live correction routes. Such routes have to be independent enough to detect a discrepancy with the target and translate it into altered production, uptake, withdrawal, or later behaviour. Call the graded relation answerability and the stronger arrangement-level capacity corrective control. This paper analyses both through a profile that records which routes constrain an output-producing arrangement and how those routes are related. Answerability comes in degrees; corrective control depends on whether the arrangement can access the target, detect a sufficiently independent and attributable discrepancy, and act on it.

Language models provide the pressure case, with the text-only profile treated as a limiting case rather than a product description. Systems built around text-trained models inherit patterns shaped by testimony, coherence constraints, and prior correction. Those patterns can supply derivative answerability where target-sensitive correction survives in the training distribution. But corrective control fails wherever the output-producing arrangement's live routes don't jointly supply the features a task requires. This predicts fluent outputs that fail when a task needs a fresh, independently informative route to the facts. Self-consistency, retrieval, tools, code execution, multimodal input, and feedback should help selectively. Such interactions test the proposed distinctions among routes. The decomposition earns its keep if, fixed in advance, it predicts performance in new combinations of routes and tasks or improves intervention choice without conceptual refitting. Surface improvement and truth-tracking improvement can come apart.

Keywords: large language models; grounding; answerability; corrective control; projectibility; truth-tracking

AI USE. The large language models Anthropic Claude Opus 4.8, Fable 5, and Opus 5; and OpenAI Codex using GPT-5.5 served as drafting and editing aids throughout the preparation of this paper. I'm responsible for all theoretical claims, arguments, errors, and interpretive choices.

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I INTRODUCTION: FROM GROUNDING TO CORRECTIVE CONTROL

Do the representations inside large language models (LLMs) connect to the world? The question is often put as a yes/no. Bender and Koller (2020) say no: a system trained on form alone, like an octopus eavesdropping on telegraph traffic, learns the statistics of language apart from what language is about. Others reply that such systems meet some conditions on reference, or that sensory grounding can be optional for thought in the first place (Chalmers, 2023; Coelho Mollo & Millière, 2026).

But that opposition only starts the inquiry. The debate descends from the symbol-grounding problem (Harnad, 1990): how can symbols, or symbol-like states, get meaning independent of the interpretations supplied by users? For LLMs, recent work has pluralized the issue. Pavlick (2023) treats grounding as a family of questions about symbols and use; Coelho Mollo and Millière (2026) distinguish kinds of grounding; and Chalmers (2023) separates sensory grounding from thought.

The recent literature has answered sceptical arguments one at a time, principally on the side of content. On reference, natural histories of a language model's inputs can secure it (Mandelkern & Linzen, 2024), while intentions are the wrong place to look (Pepp, 2025). On linguistic competence and meaning, the argument from communicative intention fails at both premises (Attah, 2025), and facts about linguistic practice can determine meaning even if the facts that would determine mental content don't (Grindrod, 2024). At the level of the proposed target, meaning can be grounded in a corpus rather than in the world (Havlik, 2024). Ruyant (2026) reaches the further conclusion that what such systems represent is a class of appropriate linguistic production rather than the world.

I contest none of those claims, and my argument doesn't require their authors to be mistaken. Some of those authors discuss misrepresentation, error, and repair, but none supplies an account of the live, task-indexed routes through which a fresh discrepancy can be detected and translated into a target-improving revision.¹ Establishing content and establishing the capacity to detect and repair fresh errors are different achievements, and the second doesn't follow from the first.

Distinguishing kinds of grounding still leaves open which differences in epistemic architecture explain patterned truth-relevant success and failure. Truth-relevant success depends on STABILIZING ROUTES² such as perception, testimony and records, coherence constraints, measurement, intervention, checking and revision, and feedback from practical success or failure. Some bring target information into production; others integrate, check, or correct what's already there.

AN ARRANGEMENT is a language model together with whatever prompts, records, tools, sensors, checkers, or reviewers are live for a particular target and task. Its PROFILE records which routes constrain it and how those routes are coupled: which can check, calibrate, or correct which others. An output is ANSWERABLE to a target to the extent that discrepancies can affect what the arrangement produces, accepts, corrects, or withdraws. The arrangement has CORRECTIVE CONTROL when live routes let it detect and repair those discrepancies.

The arrangement bears the profile and corrective control; the language model or output alone doesn't. Taking the surrounding practice without limit would instead confer the epistemic credit of

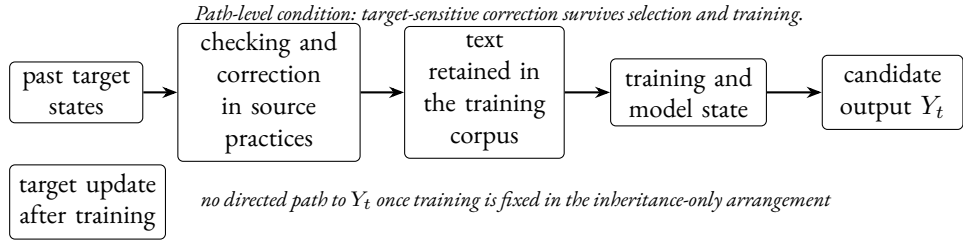
¹Coelho Mollo and Millière (2026) build the possibility of misrepresentation into their account, and Attah (2025) discusses conversational repair. Their treatments stop short of the live routes from discrepancy to revision at issue here, and neither is committed to denying that such routes matter.

²In Grindrod's (2024) reading of Millikan, a STABILIZING FUNCTION is instead conventional and inter-agent: it sits between speaker and hearer. It explains why a linguistic device keeps being reproduced, and he puts the notion to work on language models. A device can have a robust stabilizing function in Millikan's sense while preserving only the effects of the routes at issue here. That's arguably the text-only case.

human science on every deployment. The profile earns its keep if it predicts where interventions will and won't transfer.

LLMs matter because they separate routes that human epistemic life normally bundles together. The separation is clearest in the limiting case of an arrangement whose contact with the world remains mediated mainly by previously produced human text rather than live retrieval, perceptual inputs, execution results, or action outcomes. That case supplies a baseline, not a description of every LLM arrangement. Where training preserves the effects of target-sensitive correction in source practices, it can supply **DERIVATIVE ANSWERABILITY**. **LIVE ANSWERABILITY** adds a temporal condition: a fresh discrepancy has to be able to reach current production. Relative to the baseline, retrieval, tools, code execution, multimodal input, and action-guided feedback add live routes, reducing reliance on inherited constraints for the tasks those routes can serve. Their diagnostic value depends on which route each adds and where any resulting improvement transfers. Figure 1 sets out the difference.

(a) Historical inheritance



(b) Live target access and correction

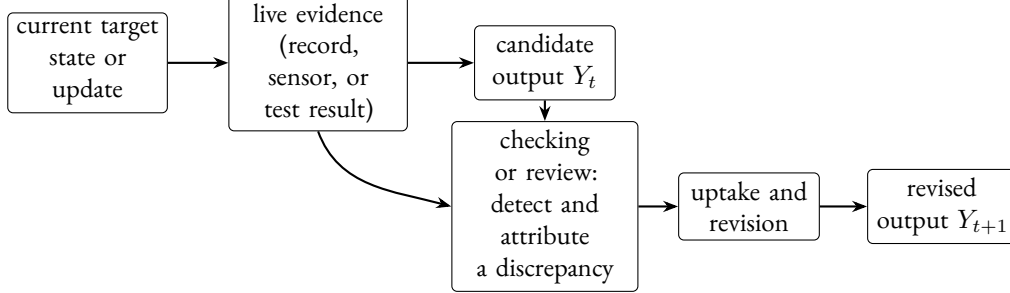


Figure 1: Historical inheritance and a live correction path. Solid arrows mark possible causal influence, not answerability or warrant. In panel (a), effects of checking and correction in source practices may survive in training text and shape the model state that produces Y_t . The path supports derivative answerability only if target-sensitive correction survives selection and training; a target update after training has no directed path to Y_t in the inheritance-only arrangement. In panel (b), current target-sensitive evidence can affect Y_t and be checked against it. A detected and attributable discrepancy can trigger uptake and revision, producing Y_{t+1} . This path supplies live answerability. It supplies corrective control only when the evidence reaches the arrangement, the check is sufficiently independent and attributable, and uptake improves the output.

An intervention should help only where it supplies a route the task needs. Self-consistency should help most where the missing support is coherence; retrieval should help most where the missing support is record access. Feedback based on human preference judgements should improve conversa-

tional fit before it improves answerability. Multimodal input should help some perceptual tasks before it helps calibrated measurement. These contrasts provide an initial test of the proposed distinctions among routes. Indiscriminate transfer would count against it.

These contrasts compare epistemic routes while leaving open what truth consists in. Correspondence, coherence, pragmatism, reliabilism, and social epistemology each identify a relation or process that can constrain representations in ways relevant to getting things right. Deflationism leaves those explanatory questions open while denying that truth itself requires a substantive reduction. The comparison concerns how these relations and processes combine within an arrangement and what follows when one is missing. It's a framework for comparison, not a reconciliation of the theories.

This paper develops that account of epistemic architecture in three steps. First, it isolates the live, task-indexed routes by which a fresh discrepancy reaches production as something to be explained in its own right rather than a corollary of content. A positive grounding verdict can settle what a system's representations mean or refer to without licensing an inference about which fresh errors a deployed arrangement can catch.

Second, it separates the profile from corrective control. Route identity and coupling determine the kind of target constraint available and where it should help. Five arrangement-level features then diagnose whether target-relevant variation can expose a sufficiently independent and attributable discrepancy and produce a target-improving revision. They also separate the capacity to correct from observed correction performance.

Third, it turns the full decomposition into predictions that can fail. Route-by-task interactions test whether the proposed routes track selective transfer, but a relevance-sensitive information account can predict them too. The stronger test is held-out transport to new cases. A decomposition fixed in advance should predict new combinations of routes, tasks, and perturbations better than a prespecified model of the task-relevant information supplied to an arrangement. It should also guide intervention choice without conceptual refitting.

Sections 2–4 define the target, organize the routes, and state the predictions licensed by the profile. Sections 5 and 6 compare the proposal with component theories and objections. Section 7 applies it to LLM arrangements and derives the empirical transfer tests.

2 THE TARGET: TRUTH-RELEVANT REPRESENTATIONAL SUCCESS

Before the profile can be applied, its target has to be fixed. This paper doesn't profile truth considered as an abstract property or as the role of the truth predicate, neither of which has perceptual, testimonial, or instrumental mechanisms as parts. It profiles TRUTH-RELEVANT REPRESENTATIONAL SUCCESS: beliefs, assertions, inscriptions, measurements, models, and inferential outputs assessed as getting things right.

That target extends beyond central TRUTHBEARERS. Beliefs and assertions can be true or false in the ordinary sense. Neighbouring cases support truth-ascription by yielding, recording, or constraining claims: a measurement, map, statistical model, or experimental display may be accurate, calibrated, or well-fitted in ways that make associated propositions true.

These cases often divide the representational labour. In a lead-exposure case, the assay reading, calibration log, database entry, and clinician's assertion do different jobs. Only the last may assert that the sample contains 3.2 mg of lead, but the others make that assertion available, checkable, and defeasible. A map can likewise preserve spatial relations that support claims about location, and a

model can constrain which inferences remain live. The object of analysis is the background through which such outputs support getting things right.

Fixing the target this way leaves a reductive definition of truth to one side. Deflationary and minimalist theories can give the truth predicate a thin logical or generalizing role while keeping a substantive causal story out of the basic theory of truth (Horwich, 1998). That point can stand. What remains to explain is why truth-relevant representational successes cluster and fail in patterned ways, and when evidence about them supports expectations about nearby cases.

The target leaves correspondence intact. Correspondence theories preserve the idea that truth involves answerability to how things are (David, 1994; Kirkham, 1992). Correspondence fixes the target of answerability; it doesn't specify the supporting routes.

ANSWERABILITY generalizes the constraint beyond central truthbearers. A representation is answerable to a target when discrepancies between it and the target can bear on its production, uptake, correction, or withdrawal. Strictly, profiles belong to arrangements, and an output is answerable relative to the arrangement that produces, maintains, and revises it. Section 7.1 identifies the relevant arrangement as the language model together with the prompt and any retrieved records, tools, sensors, checkers, reviewers, or update routes live for the target and task. For brevity, I sometimes call an output answerable without restating the arrangement that supports the relation.³

Answerability comes in degrees because routes differ. A calibrated instrument is answerable because discrepancies can appear in checks, recalibrations, and downstream failures. A coherent but disconnected arrangement is less answerable where the relevant fact has no route into production. Delayed uptake, audit, replication, practical failure, or correction of later behaviour can still connect a fact to an output. The connection is stronger when the route is specific, repeatable, and close enough to expose the relevant error; it's weaker when the route is delayed, generic, or loosely tied to the claim.

Different routes make the gradation concrete. Repeated weighing, standard weights, and failed downstream uses can expose scale drift. Records, other witnesses, and later events can defeat a mistaken report. A language model that generates a plausible citation from textual pattern alone has so far been shaped by genre, with those checks absent from the production route. Its citation may still be right, but its rightness is less supported by the route that produced it.

Code execution supplies a contrasting live route. A language model can write a function, run it, inspect a compiler error or failed test, and revise the code. A one-shot generator lacks that route even when its first answer happens to work.

The contrast survives when content is held fixed. *The sample contains 3.2 mg of lead* is strongly answerable when it's generated by a calibrated assay whose logs, standards, and downstream checks can expose error. The same string is weakly answerable when a text-only arrangement produces it because such numbers fit the genre. The truth conditions haven't changed; the route by which the claim became available has.

Pragmatist theories identify practical correction as another route to answerability. Inquiry, action, and successful practice can correct locally coherent or socially inherited representations with weak world-connection (Dewey, 1941; James, 1907). That doesn't make truth identical with what

³The lexical history of *true* is illustrative, not evidential. The *Oxford English Dictionary* records senses of loyalty, veracity, accordance with fact, accurate fit, and mechanical alignment (Oxford English Dictionary, 2026). They give the broader constraint relation historical precedent, but inferring the analysis from them would be an etymological fallacy.

works. Practical success helps arrangements become answerable to their targets.

Here, TRUTH-TRACKING has a modest sense. It names patterned support for representational success, whereas Nozick's (1981) truth-tracking account specifies subjunctive conditions on knowledge. Patterned support lets us expect some representational successes to survive checks, perturbations, and neighbouring applications better than others.

Boyd's (2021) accommodationism explains why such support can be patterned. Reference and truth depend on more than definitions in isolation; they're achievements of practices that align perceptual, instrumental, cognitive, and representational activity with relevant causal structures. That account raises a narrower diagnostic question: how strongly do those routes and practices constrain a given arrangement?

For LLMs, that diagnostic asks how much of this background a given arrangement inherits or instantiates. A text-only arrangement may inherit patterns shaped by testimony, statistical regularities, and local coherence constraints while lacking live perception, measurement, intervention, and ordinary practical correction. That profile makes truth-relevant assessment possible while predicting uneven, empirically diagnostic success.

With the target now fixed, the analysis turns to which route patterns support projection to nearby cases.

3 ROUTES AND PROFILES

Goodman's (1955) new riddle asks which predicates support PROJECTION from known to unknown cases. Here, PROJECTIBILITY concerns whether evidence about an arrangement's route profile warrants expectations about its likely success or failure on nearby tasks.

Khalidi's (2013) account provides the causal-network framework for this use of the profile. Natural categories earn their status by supporting prediction and explanation, and they do so when associated properties stand in causal relations that make some properties projectible from others.

I apply the framework to truth-relevant representational success. The framework distinguishes the evidence that warrants a projection from the world-side relations that make it reliable. Relevant routes include perception, testimony and records, coherence constraints, measurement, intervention, checking and revision, and feedback from practical success or failure. Together, these routes make representational outputs more or less answerable to the world. Khalidi's (2018) later "nodes in causal networks" formulation supplies the graph vocabulary. On his account a kind is a node, a highly connected vertex whose core properties give rise to derivative ones, and the network is what the node sits in. A profile isn't a kind in that sense. I take from him the discipline that connectivity is the proposed world-side ground of the projections, not a claim that truth-tracking is a node with a core.

Routes do different jobs. Some bring target information into the arrangement, some integrate it with existing representations, some expose discrepancies, and some return practical failure to later production. A route can do more than one of these jobs. The routes' corrective significance depends on whether, together, they let the arrangement detect and repair a fresh discrepancy.

Calling the result a profile marks two points. First, different routes can dominate in different domains. Proof norms matter more in mathematics than ordinary perception does; calibration matters more in laboratory measurement than conversational fluency does. Second, the epistemic significance often lies in the relations among routes. Testimony matters differently when it's backed by records, instruments, and correction than when it's an isolated report.

This profile view draws on Boyd (1991, 1999), who develops homeostatic property cluster theory to explain how real kinds can be projectible without essences. Boyd's (2021) accommodationism adds the detail that matters here. Perceptual skills, instruments, inferential routines, social arrangements, and representational uses each contribute when they help a practice align with relevant causal structures. The routes form a coupled pattern, not coordinate epistemic goods.

Accommodation, in this sense, is the gradual fitting of representational practice to the structures it's used to track. A scientific term, an instrument reading, or an ordinary object name becomes more reliable as perception, measurement, inference, correction, and use are adjusted together. Definitions alone don't secure the fit; practices that make error detectable and revision possible maintain it.

Consider *fever*. Feeling hot, thermometer readings, calibration standards, thresholds, records, and revised estimates of normal body temperature have adjusted one another over time. The term remains useful because these routes expose error and let the practice revise its measurements.

Applied to text-trained arrangements, the analysis remains continuous with Boyd at the explanatory level and diagnostic at the arrangement level. Accommodationism tells us why representational practices can be answerable to causal structure; the profile records how strongly the relevant routes constrain a given arrangement. A text-trained arrangement can inherit stable linguistic and testimonial patterns while its instrumental, perceptual, and practical routes remain attenuated.

In arrangements for everyday perception and basic coordination, perception, action, memory, testimony, and practical failure – the routes where error has been costly longest – have repeatedly corrected one another. A language-model arrangement that lacks perceptual and verification routes should show a corresponding failure pattern.

In arrangements for frontier science and remote historical fact, the routes may instead be sparsely linked. Error cost is delayed, diffuse, or beyond current instruments. There may be inference norms, expert testimony, and coherence constraints, but fewer shared routes of correction. Shared selection or training history, accumulated correction procedures, and measurement infrastructure are often limited there too.

Network boundaries still depend on the chosen level of abstraction. Onishi and Serpico (2022) show this for causal graphs, and Craver (2009) makes the same point for mechanism talk. Following Khalidi's (2013) distinction between epistemic purposes and non-epistemic practical or aesthetic ones, the purpose here is to represent a target domain well enough for relevant inferential, practical, or corrective success under perturbation. Connections can be sparse without dissolving an arrangement's profile relative to that purpose.

4 CORRECTIVE CONTROL AND PROJECTIBLE PREDICTIONS

Routes and their relations define the profile. The remaining projective question is when evidence about that profile warrants expectations about further truth-relevant success or failure.

Perturbation makes the declared purpose empirically testable. It distinguishes lucky fit from robust answerability. A claim based only on success in the training cases, the familiar environment, or the current conversational context supports only narrow projections. It earns stronger warrant when the representation continues to guide action, inference, measurement, and correction after conditions change.

Testing that persistence requires only a modest causal framework. Pearl (2009, 2010) distinguishes association, intervention, and counterfactuals. Observed route identities and relations de-

scribe the initial profile. Perturbation helps identify mechanism, and projectibility asks counterfactually what would hold nearby. Woodward (2001, 2003) supplies the manipulability reading: a route is explanatory when interventions on it change performance in stable, predictable ways. The argument doesn't require a fully specified structural causal model.

What matters is invariance under the declared changes. A route pattern can support bounded expectations without being actively maintained by correction. Corrective control is one way such robustness can be produced, but neither past stability nor projectibility by itself establishes live error detection and repair (Reynolds, 2026).

CORRECTIVE CONTROL adds capacities for discrepancy detection and target-improving change. An arrangement has corrective control only relative to a target and perturbation range. Its strength within those bounds depends on five features of the arrangement's corrective architecture:

1. TARGET ACCESS: target-relevant variation reaches the arrangement through an operative route.
2. DETECTABILITY: a discrepancy can be registered by that arrangement against a standard it can consult.
3. CHECKING-ROUTE INDEPENDENCE: the checking signal doesn't merely reproduce the production route's error.
4. ATTRIBUTION: the discrepancy is specific enough to guide a response.
5. EFFECTIVE UPTAKE: when a relevant discrepancy occurs, is detected, and is attributable, the arrangement can make a change early enough to reduce it within the declared horizon.

These are graded, potentially dependent dimensions; the numbering is for reference, not rank. Target access, for example, belongs to the arrangement rather than to a sensor, record, or checker considered alone: a route may be present without transmitting the relevant variation for the declared task. Likewise, a checker's errors may partly correlate with the generator's, attribution may be more or less precise, and uptake may be delayed or probabilistic.

A single architectural change can improve several features. Giving a checker access to repository state, for example, can improve target access and reduce its error correlation with the generator. This account ranks one arrangement above another when it instantiates every feature at least as fully and some more fully. Trade-offs needn't be rankable: greater checking-route independence can come with weaker attribution. Nothing here licenses a weighted score across the five.

These five features characterize corrective capacity, not rates. Effective uptake is especially easy to mistake for a rate, on which a route contributes in proportion to how often detection alters production. That conflates architecture with observed rates. How often an arrangement corrects depends on how often relevant discrepancies arise, so a near-flawless arrangement in a benign environment would score low while a poor generator triggering a mediocre checker would score high. Counting any alteration also counts alterations in the wrong direction.

Instead, effective uptake is conditional and directional. Given a discrepancy that occurs, is detected, and is attributable, can the arrangement make a change that reduces it? That's a question about architecture. What varies separately is **CONDITIONAL REPAIR PERFORMANCE**: the probability that the arrangement makes an appropriate target-improving change when a relevant discrepancy is present, considered alongside its rate of unnecessary or harmful intervention when correction isn't warranted. Detection and uptake after detection can be reported as component rates. Audited end-to-end correction and false-intervention rates provide evidence about capacity and observed performance. Raw repair frequency isn't what corrective control consists in.

Among the five, checking-route independence alone compares one route's errors with another's. It depends on where a checker came from, what it can see, what it's optimizing, and how its errors covary with the generator's. Target access, detectability, or effective uptake can be present with it or without it.

These conditions fix the relation to answerability rather than competing with it. Answerability comes in degrees, and its strength depends on how specific, repeatable, and proximate the route is. Corrective control is stronger than answerability alone: it requires both discrepancy detection and an effective route to repair. So an arrangement can be answerable in some measure while lacking corrective control over the targets at issue.

A thermometer connected to an independently calibrated drift monitor illustrates the distinction. Suppose it consults a separate standard, raises a correct alert, and every alert is ignored. Target access, detectability, checking-route independence, and attribution are present. Effective uptake is absent: no route connects the alert to what the arrangement does. Because the people ignoring it are part of the arrangement, that's a missing capacity rather than a low correction rate.

By contrast, an approval step that sometimes rejects requests shows that its intervention route is exercised. But rejection frequency isn't correction quality. Estimating conditional repair performance requires knowing which requests contained relevant discrepancies, whether those were rejected appropriately, and how often safe requests were rejected unnecessarily.

The relevant projections vary by target and task. Arrangements used in everyday perception and basic coordination often include routes that detect departures and repair them. Arrangements used in frontier science and remote historical inquiry may sustain recurrent success without comparable evidence of live correction. Text-only arrangements instantiate that combination: their outputs depend on stable textual structure while their routes for detection and repair remain weak.

Return to the lead assay. A value copied from yesterday's database entry may remain right while the sample and instrument conditions stay fixed. A value produced through a live assay, calibration log, and error-reporting practice warrants stronger expectations about nearby cases because changes in the sample or instrument can affect the result.

Survival and reproduction mark limiting biological cases of this structure. Hoffman et al. (2015) warn that adaptive success and veridical representation can come apart. Martínez (2019) argues that usefulness can drive representations toward truth under specified conditions. Usefulness becomes truth-relevant when the task, environment, and correction routes make error costly in ways that expose the relevant mismatch.

Outside biology, institutions and technologies often supply the routes. In science, law, medicine, engineering, testimony, and language-model evaluation, the relevant routes include replications, instruments, standards, records, peer challenge, and downstream practical failure. These routes make an arrangement's truth-relevant success more stable under perturbation.

Useful classifications show why *works in practice* is too coarse on its own. A classification can succeed because it tracks a real structure, because the environment has been arranged around it, or because a task rewards a proxy. Only structure-tracking, and some forms of environmental arrangement, support the kind of projection at issue here. The analysis has to identify which route supports the success.

Evidence that routes are coupled can warrant projection. A judgement based only on immediate visual appearance provides weak grounds for projection if a change in lighting or perspective, or the

addition of testimony or measurement, readily overturns it. The same judgement warrants stronger expectations within a practice where perception, action, memory, testimony, and correction have repeatedly calibrated one another.

The domain contrast introduced in Section 3 now bears on warrant. Mathematical and logical projections may rely mainly on inferential structure; scientific projections on instrumentation and replication; testimonial projections on mature institutional roles, records, and correction. No one route provides the evidential basis in every domain.

Edge cases follow from the same structure. Frontier science and remote history often rely on sparse or poorly calibrated route profiles. Genuine routes may remain, but they provide limited grounds for expecting agreement, correction, or future success.

Applied to language models, this diagnosis becomes concrete. A text-only arrangement whose inherited constraints come mainly from testimony and local coherence, with weak live routes for perception, measurement, intervention, and practical correction, should show a specific failure profile. It can produce fluent outputs that appear truth-relevant under textual constraints alone and fail under perturbations that require a live route to the target.

Tool use, retrieval, multimodal input, expert review, and action-guided feedback matter because they change the route profile. They may make projections more reliable in some domains, leave them unreliable in others, or introduce new routes to error. The predictions concern contrasts among arrangements and tasks, not a single ranking.

5 WHAT PROFILE STRUCTURE ADDS TO COMPONENT THEORIES

Profile structure contributes when it answers questions that are hard to formulate if one route is treated as the main unit of assessment. The answer turns on coupling: a history in which routes have calibrated one another can itself be evidence.

Each theory captures a local source of success. Reliabilism foregrounds reliable processes (Goldman, 1979). Coherentism foregrounds integration, with input needed for truth-conduciveness (Bonjour, 1985). Pragmatist accounts stress correction through successful practice (Dewey, 1941; James, 1907). Correspondence accounts preserve answerability to the world (David, 1994; Kirkham, 1992). Social epistemology treats testimony and institutions as genuine epistemic sources (Coady, 1992; Goldberg, 2010; Lackey, 2008).

Reliable production, coherence, answerability, action, testimony, and institutions each contribute to truth-relevant success. Their contributions become easier to distinguish when the underlying routes are separated, recombined, or moved into a new arrangement.

Profile structure adds evidence from calibration among routes: perception checked by testimony, testimony disciplined by memory and records, coherence constrained by measurement, measurement corrected by intervention, and practice corrected by failure. Call the history of those relations the arrangement's CALIBRATION HISTORY. Evidence from that history matters where it predicts current cross-route error structure or stability under perturbation. Where evidence establishes present links as causal, a network representation exposes structure that source-by-source assessment leaves in the background: the strength and history of the links between nodes.

A source's evidential value can change when the rest of the correction network changes. Past success through a locally reliable route may stop supporting projection when the route is removed from the practices that made it reliable. The WHO Surgical Safety Checklist supplies a larger-scale

case: copying the checklist without the team coordination, local adaptation, coaching, and feedback involved in effective implementation need not reproduce its effects (Armstrong et al., 2022; World Health Organization, 2009).

LLMs sharpen this point for reliabilism in particular. The relevant process is rarely just *the language model generated this text*. It may extend from the data, filtering, and feedback used to train the model, through the prompt, deployment rules, and any retrieved records or tool calls available at runtime, to checks and institutional decisions made after generation. Reinforcement learning from human feedback (RLHF) is one source of training feedback. Reliability often attaches to the process implemented by that whole arrangement. A cleanly bounded internal generator may be too narrow. Reliabilism can preserve its central insight by individuating the process at the same level at which the routes operate.

Reliability and answerability cross-cut. A route can be reliable while weakly answerable, and answerable before it's reliable. It may be reliable in a familiar environment but weakly answerable under perturbation if target-relevant changes have no route into production. Conversely, a newly calibrated instrument can be answerable before it has much of a track record: standards, logs, and downstream checks already specify how discrepancies would show up. Reliability is a statistical property of a process. Answerability is a counterfactual relation between a representation and a target, relative to an arrangement. Corrective control is an architectural property of the arrangement; an audited track record of corrections can estimate its conditional repair performance, while an ordinary accuracy record can't do so by itself.

A transfer case shows why process individuation matters. Suppose a retrieval-augmented arrangement, which retrieves external records and places them in the language model's context, scores well on source-backed factual questions about clinical guidelines. A coarse process description such as *this system answers medical questions reliably* invites transfer to neighbouring medical tasks. That coarse description doesn't warrant the transfer. The success was supported by record access, text extraction, and institutional testimony. It should transfer to nearby record-backed questions and degrade on tasks whose missing route is live measurement, patient-specific observation, calculation, or intervention.

A careful reliabilist can individuate the process more narrowly, as retrieval-backed answering under a particular profile of records, tools, and checks. That narrower individuation reproduces the profile-level decomposition. The difference is methodological: the proposal requires that decomposition before the transfer is granted.

Two cases show where the accounts come apart in practice. A newly calibrated instrument has no track record of its own, while its standards, logs, and downstream checks already fix how a discrepancy would show up. In the converse case, a long-reliable procedure keeps its record intact after the checks that supported it are quietly withdrawn.

Neither case refutes reliabilism. A reliabilist who assesses process types rather than individual track records handles both. The new instrument inherits the reliability of a type it belongs to by calibration, and withdrawing the checks changes which type is in play. The reliabilist can re-individuate after either case, and will.

Re-individuation doesn't eliminate the disagreement. Type membership has to be fixed by something, and doing so requires knowing which checks were carrying the reliability, which is what the profile supplies. The disagreement isn't about entitlement to project. It's about which variable pre-

dicts when projection fails, and this profile names one: a reliability estimate carried across a change that removes the cross-source checks should degrade, and an account that leaves that variable out will predict worse.

5.1 ADJUDICATING MECHANISM CONFLICT

When two routes disagree, say perception reports one thing and well-attested testimony another, a theory focused on one source at a time underdescribes the evidential situation. Goldman-style reliabilism can ask which process type is reliable, including conditional reliability (Goldman, 1979); social reliabilism can extend the assessed process through testimonial production (Goldberg, 2010). Those are genuine resources. The further question is how tightly these sources have been calibrated against one another in this domain.

In Khalidi's (2013) terms, causal relations among associated properties underwrite projection; here the projective inference runs from one route's output to another's.

Consider a modest conflict. A subject's visual judgement says that an object has property *P*. A calibrated instrument says that it lacks *P*. Several competent people, drawing on the same instrument tradition, say the instrument is functioning normally. The profile supplies no standing rule to trust perception, testimony, or the instrument. What matters is which sources have historically corrected which others in this region of practice.

In everyday object recognition, ordinary perception, practical feedback, memory, and testimony have long corrected one another. If I seem to see an old friend at the end of a hallway while well-placed testimony rules out her presence, the perceptual judgement can reasonably give way. Coady's (1992) discussion of perception, memory, inference, and testimony already pushes in this direction: the sources interpenetrate, and testimony can rationally undermine what presents itself as perception.

In a mature instrumental domain, the priority can reverse. Unaided perception may be less reliable than a network of independently calibrated instruments, records, expert testimony, replication, and repeatable intervention. Within such a network, a lone divergent output is defeasible evidence of local error or a boundary case. Agreement copied from a common source doesn't license the same inference; where routes are loosely connected, divergence reduces confidence without locating the error. Global source reliability is too coarse. The verdict remains empirical and fallible: both cross-route calibration and a source's assessed reliability enter the evidence.

Russell's (1907) classic objection to the coherence theory of truth made the structural point early: a false proposition can cohere with some specified set, so coherence alone can't select truth. I retain the lesson without adopting Russell's theory. Coherence constraints supply one route among several and support projection only when input and correction routes constrain them.

Semantic fragmentism gives the coherence-only case its most developed contemporary statement. Havlík (2024) argues that linguistic meanings are grounded in a corpus rather than in the world or in an internal representation of it. A corpus counts as minimal once further additions leave the grounding of its parts unchanged.

His target is meaning rather than truth, so Russell's objection doesn't transfer as a refutation. The structural parallel holds anyway. A saturated corpus fixes the meanings of its parts whether or not those parts are true of anything, and nothing inside it registers drift. Havlík grants that language models are sensitive to their training data, and that errors in it degrade what they do.

Errors enter by that route, and the account supplies none by which they leave.

Bayesian work on witness independence and coherence establishes the evidential point: agreement confirms more when sources are sufficiently independent than when they share a source or an error process (Bovens & Hartmann, 2003; Olsson, 2005). This account uses the point differently. The relevant structure includes current cross-route dependence and whatever calibration history predicts its stability. That structure helps individuate arrangements, assess transfer, and evaluate bodies of testimony.

The difference predicts failures of transfer. Suppose a language model, instrument, or witness has been reliable in a domain where perception, records, and measurement have repeatedly corrected one another. Moving the source into a neighbouring domain that lacks that calibration history should degrade performance, especially under perturbations that expose the absence of those checks. A source-by-source assessment that carries the prior reliability score across the move will predict worse.

5.2 AN AUXILIARY CROSS-DOMAIN PREDICTION

A network view yields one auxiliary prediction: persistent disagreement should cluster where independently characterized cross-route connections are sparse. Khalidi (2018) distinguishes the generality and variety of the projections a predicate supports. Here, jointly disciplined routes should support broader transfer than routes that seldom check one another.

Connectivity has to be characterized before present agreement is observed and relative to a fixed representational aim. Relevant evidence includes shared selection or training history, accumulated correction procedures, and measurement infrastructure. Recurring error signals can join perception, naming, action, and memory; calibration routines, records, replications, and intervention can join instruments and expert reports. These relations create occasions for one route to expose another's errors. Longevity or consensus alone doesn't.

Those relations are defeasible indicators, not criteria. Stakes, incentives, ideology, small samples, strategic ambiguity, and value-ladenness can also sustain disagreement. The auxiliary prediction is tighter agreement where routes are densely linked by correction and looser agreement where those links are sparse, with frontier science and remote historical fact as expected edge cases. Failure would weaken this cross-domain extension, not the distinction between grounding and corrective control. Whether the pattern extends to normative disagreement remains open.

The auxiliary prediction extends the narrower claim that a route's evidential value changes with its calibration background. Language-model arrangements make that dependence especially visible: they can inherit textual routes and coherence constraints while perceptual and verification links remain attenuated.

6 OBJECTIONS

6.1 INTEREST-RELATIVITY

One objection questions the profile's objectivity. Onishi and Serpico (2022) argue that Khalidi's causal-network account inherits the interest-relativity Craver (2009) finds in mechanistic accounts. Which causal graph is drawn, at what level of abstraction, and with which boundaries all answer to explanatory interest. The network never fixes a unique, interest-free carving of kinds. If that's right, this profile of truth-relevant success is only as objective as our purposes.

The objection assumes that an account of kinds owes an interest-free carving. Khalidi's (2013) account doesn't. He treats the sciences as studying a jumble of partially overlapping domains, each with its own repertoire of categories and methods, and holds that where two schemes sort members

into categories that don't coincide there's often no ground for ruling either scheme non-natural. Isotopic and isobaric nuclides carve the same atoms differently. Neither is the privileged carving, and the projections each supports hold up.

The reply has a metaontological cousin. On an easy approach to ontology, existence questions are settled by trivial inferences licensed by a concept's rules of use, which moves the substantive work from whether the entities exist to which concepts we should employ and why (Thomasson, 2015). I take no stand on that general thesis. The useful point is the order of business. First ask which profile concept to adopt for which inferential and corrective purpose. Then ask whether that concept's declared projections succeed. The second isn't answered by the first; that separation keeps the proposal from reducing to a report of our interests.

Purpose-relativity is a condition of application. What matters is whether the purpose is declared before the verdicts are collected, and whether the claims it licenses can fail. Held to the declared representational purpose, two claims follow: arrangements missing perceptual and checking routes should show the predicted failure pattern, and connectivity should track selection history, accumulated correction procedures, and measurement infrastructure.

A sharper objection would succeed if the purpose were chosen after the verdicts to secure them. The account would then be unfalsifiable. Characterizing connectivity through marks established before present agreement is checked blocks that move. Declaring the interest is what makes this account refutable, not what excuses it.

6.2 RELABELING AND ABSORPTION

A second objection says that this account will be absorbed by the component theories. Reliabilism can already conditionalize process reliability. Social epistemology can extend the assessed process through testimony, records, and institutions. Coherentism can demand input constraints. On this objection, this account redescribes familiar epistemology in causal-network language.

Component theories often have resources to mention coupling. Still, the profile requires the arrangement's current route structure, together with any calibration history bearing on its stability under perturbation, to be specified before transfer is projected. A careful reliabilist can reconstruct the same grain after the fact; the divergence is predictive rather than normative. This profile names the variable whose absence predicts degraded transfer, and an account that ignores it makes worse predictions.

At arrangement scale, that move raises reliabilism's generality problem (Conee & Feldman, 1998). Pretraining, filtering, RLHF, retrieval, tools, prompts, and institutional uptake can be carved into many process types. If the reliabilist uses selection history, accumulated correction procedures, measurement infrastructure, and architectural facts to choose the projectible carving, then the profile supplies the individuation theory.

The profile complements reliabilism, coherentism, correspondence, pragmatism, and social epistemology. It says when claims based on their local routes support projection, when those routes pull apart, and what failures should follow.

6.3 DERIVATIVE TESTIMONY

A third objection concerns testimony. Testimony is an interpersonal epistemic practice involving assertion, responsibility, trust, uptake, and defeaters (Coady, 1992; Goldberg, 2010; Lackey, 2008). If so, giving a language model testimonial standing because it was trained on human text alone misleads.

Recent work on artificial intelligence testimony raises the same role question from the other direction: conversational systems strain anthropocentric theories of testimony while still differing from ordinary human testifiers (Freiman, 2024).

That objection leaves a weaker claim intact. A text-trained language model inherits artifacts of testimonial practices: reports, explanations, records, instructions, arguments, and corrections left in text. That history may carry epistemic support into present production because many such texts were produced inside practices already partly accommodated to the world.

Training alone also leaves the language model outside the current testimonial exchange: it has neither the role of a normal hearer who can trust a speaker under standing norms of assertion nor the role of a speaker who takes responsibility for what's asserted. That derivative status explains both the reach and the fragility of text-only arrangements.

A stronger version remains. Causal descent from testimonial practice isn't current involvement in it, and something has to survive corpus selection, tokenization, optimization, and generation for epistemic support rather than mere influence to be inherited. The inheritance needs a preservation condition.

The distributional character of the inheritance suggests one. Training should most reliably preserve whatever shapes the distribution of forms: the shape of a practice, its characteristic moves, and regularities across many reports. What a particular passage carried rather than the pattern should be preserved only sporadically. Information about source identity, provenance, and defeaters should be inherited less systematically and recovered less reliably, which predicts particular fragility in verbatim anchors. The contrast is one of reliability, not of presence and absence.

6.4 OUTPUTS AS FICTION

A fourth objection denies that the outputs are the right kind of thing to assess. Ruyant (2026) applies accounts of epistemic representation from philosophy of science and concludes that a language model represents a constructed class of appropriate linguistic production rather than the world. Its makers' norms fix that class. Conversation with a chatbot is then a fiction, and its outputs are "props in a game of make-believe". Auditing props for answerability would be a category error.

The argument reaches a fictionalist conclusion independently of the grounding debate, and it states its own scope: Ruyant's generalization holds "to the extent that" such systems aren't trained on direct worldly inputs, which is a claim about routes in another vocabulary.

Ruyant supplies the reply himself. He grants that fictions can be performative, since generated text becomes student dissertations, professional emails, computer programs, news and scientific articles once a human appropriates it. His analysis stops there, and the ethical conclusion he draws next, that deployment promotes applications for which such systems are "fundamentally untrustworthy", needs an answerability notion his account doesn't supply.

Appropriation doesn't have to convert one kind of object into another. One string can be correct in the game, false about the world, effective as an artifact, and something a user is answerable for, all at once. Those are different correctness targets, and the analysis is already target-indexed.

So the question isn't whether the output is still a prop. It's which target is being assessed, and which routes were live for it: the ones active when the text was produced, and the ones the appropriating practice adds. An arrangement that takes a draft into a practice with measurement, review, and correction has a different profile against the world as target from one that merely files the same

draft. The string remains the same, as does the prop in the make-believe. That extends the fictionalist analysis rather than refuting it.

6.5 NON-EMPIRICAL DOMAINS

A fifth objection comes from mathematics, logic, ethics, fiction, and law. In these domains, perception, measurement, intervention, and practical feedback seem peripheral or misplaced. If this profile is built around world-contact, it may overfit ordinary empirical truth.

The objection applies only under a wider scope than this paper claims. The target here is world-directed representational success, where getting it right means getting some independently variable matter right, and where the LLM argument lives. The mathematical case also marks that boundary: proof and formal checking supply detection and correction of their own, so an arrangement can be under corrective control there with no perceptual or instrumental route at all. These non-empirical instances favour the architecture; this paper doesn't characterize them.

Law, ethics, and fiction remain possible extensions. Institutional procedure, role differentiation, precedent, and contestation may supply ways of detecting and responding to mismatches, but establishing that would require case work beyond the present empirical argument.

6.6 DEFLATIONISM AND PLURALISM

Finally, deflationists and alethic pluralists may object from opposite directions. Deflationists may say that truth has no deep nature for this profile to explain (Horwich, 1998). Pluralists may say that different domains involve different ways of being true, so a single profile is artificial.

Both objections fix the ambition: this paper explains projectible patterns among truth-relevant representational successes. That target is compatible with a thin account of the truth predicate and with domain pluralism, since different routes can be privileged in different regions.

With the scope fixed, the LLM case has to supply testable predictions.

7 THE LLM CASE: INHERITED STRUCTURE AND LIVE ROUTES

7.1 THE ARRANGEMENT AND ITS ROUTES

The relevant arrangement is whatever produces the output, indexed to a target, a task, and the routes live for that task. It may include the language model, the prompt, retrieved records, tools, sensors, checkers, human reviewers, and update routes.

Retrieval belongs to the arrangement when it runs, not to the language model. Human appropriation changes the arrangement rather than adding a gloss to a fixed one. The effects of training history belong to the arrangement's causal history. A nominally multimodal or tool-using product yields a text-only arrangement whenever those routes sit idle, which is why the same product earns different verdicts on different work.

Profiles vary with the arrangement. Arrangements centred on a pretrained or instruction-tuned language model, or augmented with retrieval, tools, multimodal input, or robotic embodiment, have different route structures. The comparison is interventionist: change the routes, and the expected successes and failures should change too.

Two questions organize the LLM case. What constraints arrived through training? Which routes are live now? Training can shape production without giving the arrangement present access to the source practice. A live route can transmit a new record, measurement, or check, and it may also

support correction. A purported route that supplies no target-relevant constraint is decorative: the signs of experiment, source use, or measurement remain while the constraint tends to zero.

Live correction also has a horizon. When a correct, attributable counterexample exposes a relevant discrepancy and causes a revision that reduces it, the arrangement satisfies effective uptake within the session and nothing beyond it. The weights don't move, the next session starts where the last one began, and no other user benefits. The arrangement has corrective control for one conversation, not for the policy deployed in later conversations.

The profile identifies which routes constrain the arrangement and how they're related. The five features listed in Section 4 diagnose corrective control within that profile. A live measurement can supply target access, register and attribute a discrepancy, and drive revision, contributing to several features at once. Corrective control remains a capacity of the arrangement.

Training can supply derivative answerability, but only on a condition. Discrepancies already registered in the source practices have to shape present production through the training history. Causal descent isn't enough by itself. Dense propaganda can leave distributional traces as persistent as those of a measurement literature. A language model trained on it can inherit the shape of a practice whose strongest corrective pressures track audience response, party discipline, or persuasive success rather than the empirical target at issue. What has to survive the training history is the effect of corrections made against that target rather than against audience response, rater approval, or a party line.

Three things come apart here: causal ancestry, which any corpus included in training supplies; derivative answerability, which requires target-sensitive inherited correction; and live answerability, which requires a route from a fresh discrepancy to current production (Figure 1).

Derivative answerability by itself never supplies live answerability. Once the training history is fixed, a discrepancy arising after it has no way to reach production. Training against verifiable rewards tests this boundary. When a language model is optimized against unit tests, a proof checker, or execution results, a discrepancy between output and an independently specified target is detected and does alter production. All five features can be instantiated while the verifier runs, to the extent that it's adequate to the target, sufficiently independent of the generator, and attributable in its signal. Tests can be incomplete, attribution coarse, and verifier errors correlated with the generator's.

The training loop has then exercised corrective control over model outputs; the deployed arrangement isn't thereby correctable. The loop supplies that control only over the targets its verifier covers and only while it runs. A language model that has learned arithmetic against a checker still can't tell you whether today's figure is right. Verifiable rewards can strengthen derivative answerability, since the corrections they encode were made against an independently specified target rather than against approval alone. They don't make it live.

A live route can supply target access. It supports corrective control only if the arrangement can detect and attribute a discrepancy, receive a sufficiently independent signal, and act on it. A merely decorative invocation resembles such a route while supplying none of these features. Checking-route independence depends on the relation between routes, not on one route in isolation.

Corrective control doesn't extend beyond the arrangement's live routes, and it's bounded target by target. A user who checks a claim can supply corrective capacity for that claim. How often users check fixes the route's exercise frequency across claims; how often checking produces an appropriate target-improving change when a relevant discrepancy is present, and how often it intervenes harmfully when correction isn't warranted, determine conditional repair performance. A routinely

bypassed checking step adds nominal capacity but little exercise. So the interesting case isn't an arrangement with no corrective control anywhere, which is rare, but the ordinary one in which an arrangement is asked about targets that none of its live routes can independently check. Grounding is untouched by that boundary. Corrective control stops at it.

These causal distinctions explain several contrasts. Inherited structure helps explain why a text-only arrangement can handle a domain whose records are dense and stable. Live retrieval and tool routes explain why deployment-time access can change performance. Target-sensitive feedback supplies corrective capacity only when it can change behaviour in ways tied to the relevant error. These routes and their relations are characterized before the performance test from the training objective, data provenance, filtering history, runtime information flow, tool-call logs, sensor access, and update mechanisms. Performance on held-out cases then tests whether the profile supports the declared projections.

The limiting failure is decorative: the language of measurement without measurement; the language of citation without a record.

7.2 ROUTE-SPECIFIC INTERVENTIONS

Retrieval augmentation queries an external corpus or search system and places retrieved material into the language model's context; it adds record access when it works. Its characteristic failures are diagnostic rather than incidental: a stale index supplies a record of the wrong time, a chunk boundary supplies half a record, and an embedding match on wording rather than content supplies a passage that resembles the answer. Each leaves the archival route nominally present and materially absent: decorative retrieval implemented by a real mechanism rather than produced by a turn of phrase.

Tool use routes a subtask through a calculator, database, code interpreter, application programming interface, or instrument; it adds calculation or measurement only when the right tool is called and its result is integrated. When execution results guide revision, tool use adds a corrective loop.

Self-consistency prompting samples multiple completions or reasoning paths and privileges convergence among them; it strengthens internal convergence. An external check requires a route to the relevant facts whose signal isn't generated by the same convergence process.

Instruction tuning and RLHF change a language model's policy through demonstrations, preferences, benchmarks, or evaluations; they add a social or evaluative correction route whose target depends on the feedback.⁴

Multimodal input supplies image, audio, video, or sensor information; by itself, it adds perceptual input. Calibrated measurement is a separate route.

Multimodality matters because it seems to answer Harnad's (1990) sensorimotor prescription. A finer-grained diagnosis separates perception from action, correction, and use. An arrangement built around a vision-language model trained on image-caption corpora inherits perceptual structure only through selected, framed, and captioned images. A deployment-time image input adds a live perceptual route; calibrated measurement, intervention, and action-guided correction remain separate routes.

⁴Preference signals have factual targets. Rater approval of affirmation, deference, or confident presentation is a fact about a population and interaction setting, and RLHF can make the resulting arrangement's behaviour strongly answerable to that fact. The failure mode is target mismatch: the policy can track the rater distribution accurately while its assertions are assessed against a different, object-level target. Sycophancy is the clean case, where an arrangement accurately tracks the wrong target.

Multimodal input should improve performance unevenly. It should reduce some failures in object classification and leave measurement- and intervention-dependent tasks fragile. It should also leave room for object hallucination, where generated descriptions include objects not present in the image (Y. Li et al., 2023). That pattern is what this account expects when target access is strengthened and the correction routes are left as they were.

To test this view, hold the topic, answer format, and apparent difficulty fixed, and vary the missing route. Gains should follow the added routes. Retrieval should improve source-backed recall more than novel measurement or action-sensitive causal judgement. Self-consistency should improve coherence-sensitive reasoning more than claims that need a fresh route to the facts. Multimodal input should improve perceptual classification before it improves calibrated measurement. If gains transfer evenly across those contrasts, that would be evidence that the proposed distinctions among routes are wrong.

A deployed product can combine several of these interventions. A chat system can be instruction-tuned, retrieval-augmented, partly tool-using, and multimodal. Its profile is indexed to whichever routes are live in the task being assessed.

Which profile counts as the baseline depends on the task. The text-only case gives the cleanest limiting test: it's task-relative rather than a description of a product. Arrangements built around language models trained on interleaved image and text, on code corpora, or against verifiers include more varied routes. When retrieval, tools, or multimodal routes are idle, inappropriate, or unintegrated, the arrangement has the text-only profile for that task.

The limiting case still inherits reports written by perceivers, explanations corrected by teachers, institutional records, and descriptions shaped by instruments. It can support genuine epistemic achievements while occupying a different role from a speaker or hearer in a testimonial exchange.

Table 1 summarizes the architecture-sensitive predictions. The rows are profiles rather than a scale from bad to good.

Arrangement type	Principal architectural resources	Directional prediction
Text-only arrangement with a pretrained language model	Training-distribution regularities and local coherence	Strong where dense, stable text is a good proxy; fragile for recent, obscure, perceptual, or measurement-dependent facts.
Instruction-tuned or RLHF arrangement	Rater feedback, demonstrations, and benchmark evaluation	Better conversational fit and some error reduction; independently informative checking routes still have to be supplied.
Retrieval-augmented arrangement	Live access to records and institutional sources	Better on updated or source-backed recall; remaining fragility where the missing route is experiment, perception, or intervention.
Tool-using or coding arrangement	Calculators, databases, code execution, instruments, or application programming interfaces	Better where the right tool is called and interpreted; strongest when runtime or test results guide revision; new failures from tool choice, stale data, and test-as-proxy effects.
Multimodal arrangement	Perceptual input in image, audio, video, or sensor form	Better for some perceptually anchored tasks; calibrated measurement and practical correction remain separate routes.
Action-guided or expert-reviewed arrangement	Practical feedback, review, and downstream failure signals	Stronger corrective capacity where feedback is timely and domain-competent; weaker where review is sparse or proxy-based.

Table 1: Architecture-sensitive profile predictions.

7.3 INHERITED STRUCTURE AND MISSING ANCHORS

Textual inheritance explains both the competence and the fragility of text-only arrangements. Human text is saturated with world-directed practices of looking, measuring, arguing, repairing, citing, and correcting. The deployed arrangement usually remains outside those practices when it produces a fresh answer. It draws on their products with attenuated access to their present checks.

Text-only training can still produce more structure than a bare *textual residue* slogan suggests. The generative pretrained transformer model Othello-GPT is a challenge case: a sequence model trained to predict legal moves in a board game developed an internal representation of board state, and interventions on that representation affected its outputs (K. Li et al., 2023). The result shows that inherited constraint can be strong when the target state is densely and systematically encoded in the sequence.

Othello provides only a limited basis for a broader claim. The game is finite, fully observable, and rule-governed; its legal moves are a deterministic function of a state recoverable from the move sequence. The training signal contains the target state entirely. That’s why the result is both striking and unrepresentative. Nothing follows about domains whose relevant state is absent from the record,

which is the class this argument concerns.

Internal world-model evidence should be strongest where the relevant state variables are encoded in the sequence and the loss function penalizes violations of their structure. It should weaken when success depends on variables absent from the sequence, live measurement, changing local conditions, or intervention in the target system. Successful transfer across those boundaries would count against this diagnosis.

Alongside textual inheritance, coherence strongly constrains these arrangements. Training and decoding reward genre fit, local well-formedness, smooth inference, and continuation from context. Those pressures explain why the language models in these arrangements can display impressive formal linguistic competence, in the sense separated from functional world use by Mahowald et al. (2024). They also explain the vulnerability. Coherence can make an answer internally well-formed while adding no independently informative route by which the relevant facts constrain it.

In the baseline profile, perception, measurement, intervention, and practical correction enter weakly through text produced by others. A language model can generate a plausible lab result, legal summary, local observation, or bibliographic claim because similar text exists. Present production remains unconstrained by an experiment, court record, observation, or catalogue entry.

Textual inheritance is reliably distributional and only contingently archival. Training can and does preserve particular strings, names, and sources, so this claim isn't that token-bound information never survives. It's that nothing preserves it under a general relation of record identity, provenance, and current checkability. Training preserves redundantly represented patterns more reliably than material carried by a single passage. A text-only language model can encode statistical regularities in testimonial practice. The resulting arrangement can handle a domain whose records are dense and stable while fabricating the verbatim anchors (quotations, identifiers, author lists) on which the practice's correction routines depend. The same distinction predicts that these failures should be especially retrieval-sensitive: retrieval adds the archival route that distributional inheritance lacks.

Hallucination plays a more specific role here. The LLM literature treats it as heterogeneous (Huang et al., 2025). The narrower claim is that text-only arrangements should be especially vulnerable to fluent, coherent, genre-appropriate outputs that fail when the task requires a fresh or independently informative route to the facts. Those failures should be common for obscure sources, recent events, local observations, novel measurements, fine-grained quantities, and action-sensitive causal claims. They should be rarer where dense, stable, redundant text is a good proxy.

Frankfurtian accounts of bullshit (Frankfurt, 1986) diagnose LLM output differently. Hicks et al. (2024) apply that diagnosis to ChatGPT-style systems, arguing that their outputs are produced with indifference to truth. On the Frankfurtian diagnosis, the system doesn't care where the facts are; on this diagnosis, the relevant facts have no route into production or correction. That missing route yields comparative predictions across retrieval, tools, multimodality, and action-guided feedback. This diagnosis asks for a route, not a better attitude.

Mitigation results should be interpreted using the same distinction. Retrieval can reduce hallucination in conversation (Shuster et al., 2021); that's evidence about record access. Perception, experiment, and intervention require further routes. Tool use and multimodality make parallel changes. They add live routes where the tool, sensor, or modality is live in the task; otherwise the new vocabulary of tools or perception can remain decorative.

Action-guided arrangements can substantially expand corrective control when the feedback is

timely, attributable, and tied to the relevant failure. The code-execution case from Section 2 is the clean artificial example. An arrangement that runs generated code, observes a compiler error, traceback, failing test, or successful execution, and revises against that result has a more direct corrective route than one rewarded for plausible-looking code. A robot that breaks a glass, misses a target, or fails a laboratory protocol can receive similar correction unavailable to a text-only arrangement. A deployed chatbot that receives delayed approval, engagement metrics, or sparse user edits has a slower, less attributable corrective route. The label *feedback* hides that difference unless the route is specified.

7.4 WHAT THE CORRECTION ROUTES CORRECT

Instruction tuning, RLHF, benchmark evaluation, red-teaming, and expert review function as correction routes, and they should be classified by what they correct. Their immediate targets often include helpfulness, harmlessness, preference satisfaction, conversational fit, and benchmark success. Those constraints can reduce some errors and can also reward answer-shaped behaviour when users or raters prefer confidence, agreement, or plausible explanation.

The verifier case also separates two forms of post-training. Rewards based on a rater's approval target that approval; verifier-based rewards target an independently specified criterion, subject to the verifier's adequacy. The label *post-training* covers both, but they build different profiles. Treating checker-based reward as mere rater approval understates what it supplies; treating rater approval as truth-directed feedback overstates it. For this analysis, what matters is not the amount of post-training but the targets against which its rewards were computed.

Work on sycophancy shows the risk of treating feedback based on human preference judgements as truth-directed on its own (Shapira et al., 2026). These processes strengthen some social and evaluative constraints while leaving routes through perception, measurement, and intervention indirect or absent.

Accounts of fine-tuning disagree because they seek to explain different things. Grindrod (2024) treats it as a relatively marginal procedure that can't be what makes the difference between intentionality and its absence, since language models produce apparently meaningful text without it. Ruyant (2026) treats it as constitutive of what the system represents, because the norms of appropriateness a language model is tuned toward are what fix its target.

Fine-tuning can be constitutive of appropriateness for a purpose, the phenomenon Ruyant seeks to explain, because norm setters fix that purpose. It can remain marginal to whether outputs are meaningful in Grindrod's sense. Neither verdict settles target-directed correction. Rater approval supplies that only when the judgement is informed by a target-sensitive route: fact-checking against records, tool results, or action outcomes rather than approval alone. Treating all three questions as one quantity called *grounding* makes the disagreement look unresolvable.

7.5 WHAT GROUNDING LEAVES OPEN

Coelho Mollo and Millière (2026) present the closest neighbouring challenge. They distinguish kinds of grounding and argue that some LLM states can satisfy causal-historical or teleosemantic conditions for referential grounding. Their conclusion isn't in dispute. The further explanandum is the architecture that supports answerability and correction.

Work on neural word embeddings pushes in a similar direction: learned representations may have genuine, though limited, contents fixed by the functions they serve inside the system (Mallory, 2026).

Teleosemantic arguments of this kind fit the baseline text-only case. The reply concedes content and asks which routes constrain its present use.

Training text carries correction procedures, defeaters, and records, and a language model can learn to reproduce some of them. What inheritance alone doesn't deliver is a live route by which a fresh discrepancy reaches production and changes it.

Referential grounding covers one dimension of world-connection. Answerability depends on a wider profile of routes, including correction, measurement, intervention, perception, and practical feedback. An LLM state might be referentially grounded in their sense even as the deployed arrangement has a fragile truth-tracking profile for tasks requiring independent checking.

Mandelkern and Linzen (2024) illustrate the dissociation. They close with Izzy, isolated from birth in a sensory-deprivation chamber and reaching the world only through a text screen, and they argue that externalism dissolves the puzzle about how his words refer. Izzy's answerability is nonetheless confined to the testimonial and inferential routes his screen supplies, with perception, measurement, and intervention absent. Secure reference alongside a restricted profile is the dissociation at issue.

Pepp raises a stronger version (Pepp, 2025). She suggests that such systems may contact objects through their exchanges with users, coming into contact with whatever shaped a user's input. The inherited-live distinction splits that case. A live image-input route supplies perceptual access. Contact with the objects that shaped a user's text is inherited: the perceiving was done by someone else, and nothing in the exchange reports back when the user was wrong.

Bender and Koller's octopus and Mandelkern and Linzen's Izzy support opposing verdicts on reference. Their shared limit is architectural. In each case, linguistic traffic is the principal route, so a discrepancy that never enters that traffic can't alter present production. What isn't on the wire can't correct what is.

7.6 CORRECTIVE CONTROL IN DEPLOYMENT

A deployment case isolates what adding a checker changes. Anthropic (2026) describes a coding agent that routes each proposed action through a classifier that blocks operations it judges irreversible, destructive, or aimed outside the working environment. The case concerns safety control rather than factual truth directly. Its role here is to isolate the architecture by which a separate discrepancy signal becomes behaviourally effective.

When its policy judgement is correct, the separate classifier can supply detection and effective gating where the acting language model registers nothing itself. Adding it can give the arrangement detectability and effective uptake through a second route rather than through introspection.

Separation alone doesn't secure checking-route independence. An external checker can reproduce the generator's errors, while the generator given independently sourced environmental information may not. Relevant dimensions include development provenance, the environmental information available to the checker, objectives, and conditional error structure. The reported hardening gave the classifier more environmental information, including repository visibility and git state. Whether that change decorrelated its errors wasn't measured, and lineage alone wouldn't settle the question. Measuring it requires conditional error rates for checker and generator on a shared held-out set.

The deployment figures instantiate the earlier distinction between exercise and correction quality. Users reject 3% of individual permission requests against 39% of plans put to them for approval. The permission route isn't decorative outright, since a rejection can alter the action. But the 3% fig-

ure records rejection frequency, not conditional repair performance; without audited rates of unsafe proposals and false blocks, that performance is unidentified.⁵

7.7 EMPIRICAL TRANSFER

Route-specific transfer provides a direct test. Improvements in one route should remain limited where a task depends on a different route. If perception, measurement, retrieval, tool use, expert correction, or action feedback is confirmed to be active and reaches production but fails to alter the expected contrast, the proposed decomposition has misdescribed the mechanisms. If text-only arrangements remain mainly dependent on inherited textual accommodation and local coherence, the same vulnerability should persist even as fluency improves.

A clinical test makes this concrete.⁶ One item set asks for recommendations recoverable from guidelines or institutional records. A paired set uses the same disease area and answer format and requires a patient-specific lab value, observed sign, dose calculation, or intervention-sensitive causal judgement. This profile predicts a retrieval gain on the record-recoverable set and little transfer to the patient-specific set unless the needed measurement or tool route is supplied. Retrieval alone producing the same gain on both sets would count against the proposed distinction between routes.

The implementation is crossed rather than merely matched: closely related item families vary whether a route is available and whether the task requires it. Substantive correctness, scored blind across both item types, is the primary outcome. Online Resource 1 specifies the estimand, controls, secondary outcome, worked decision threshold, and interpretation of uncertainty. The main test is the route-by-task interaction declared before outcomes are observed.

One interaction is weaker than two. Retrieval should help the record-dependent items, and a measurement tool should help the measurement-dependent ones, so crossing both interventions against both task types tests a double dissociation. That is, each route should confer a selective advantage on the task that needs it. The contrast tests whether the proposed routes track selective transfer, but the interaction isn't unique to this profile. A relevance-sensitive information account can predict the same crossing by treating records and measurements as different task-specific information.

Coverage has three readings. Input coverage concerns task-relevant content accessible at a declared architectural boundary before the focal component produces its signal. Depending on that boundary, the inputs can include a prompt, retrieved passage, database response, sensor output, or candidate answer presented to a checker. The boundary is declared and relevance coded against a gold-standard task analysis before system outputs are observed. Input coverage is a restricted empirical comparator: two arrangements can receive the same task information while differing in what they do with it.

Conditional coverage concerns what one route adds given what the other routes have supplied. It can represent the greater value of a checker whose errors don't reproduce the generator's. Effective

⁵ The same users and product appear in two approval arrangements. The contrast is consistent with route-specific non-exercise, though it doesn't isolate the route from interface burden, action granularity, or perceived stakes. A controlled study points the same way, with human testers catching 13.6% of planted dangerous commands against the classifier's 89%, though its participants worked in a purpose-built environment and knew they were being evaluated (Anthropic, 2026). Both figures are a vendor's self-report about its own product and haven't been independently audited. In this paper about independent checking, the evidence comes through a reporting arrangement with a profile of its own. The figures are used here to illustrate the availability and exercise gap, not to measure it.

⁶ The design parallels validity argumentation in measurement: inferences from performance are licensed relative to a purpose and tested by varying method while holding the target construct fixed (Kane, 2013; Messick, 1989).

coverage includes whatever information can influence the final output given the architecture, provenance, checking relations, and uptake policy. It summarizes the usable information supplied by the whole arrangement without decomposing how it became usable.

The clearest competing account uses input coverage. Conditional coverage remains an empirical comparator when its measure is fixed in advance, but it already incorporates checking-route independence and joint error structure. Effective coverage includes nearly the whole profile. Used without fixed application conditions, it can fit any result post hoc by counting each helpful route as an increase in coverage. It then summarizes the profile's result rather than competing as a mechanism. Effective coverage may serve coarse retrospective prediction while this profile serves intervention, audit, and transfer. Nothing here requires one uniquely correct description for both purposes.

This yields an identification limit. For any mapping from profile features and environments to performance, an unrestricted coverage variable can be defined to reproduce that mapping. Behavioural results alone can't identify which architecture produced the usable information. Lineage, information flow, conditional error structure, and update routes supply the additional evidence. The empirical comparison needs fixed application conditions specified before the outcomes are observed.

The empirical burden falls on this profile's full decomposition. Route identity and coupling specify what kind of target constraint is available. The five arrangement-level corrective features diagnose whether target-relevant variation can reach the arrangement, yield a detectable and independently informative discrepancy, be attributed, and alter behaviour in a target-improving direction. Both components are fixed from architecture and history before performance is measured.

For the projectibility claim, compare a prespecified input-coverage model with a multilevel predictive model of the full decomposition. The decomposition model can include route identities, task requirements, cross-route relations, separate measurements or indicators of the five features, and theoretically declared interactions as distinct terms; it needn't assign the profile a weighted score. Estimate the models on some combinations of routes, tasks, and perturbations and evaluate predictive accuracy and transport on held-out combinations without reclassifying routes or redefining features after observing the outcomes. Probabilistic calibration is an additional criterion only if the models issue graded forecasts. The decomposition loses the projectibility claim if input coverage predicts the held-out cases equally well.

For the practical claim, the decomposition has to improve diagnosis or the choice among retrieval, measurement, independent checking, and enforced uptake beyond whatever predictive gain it supplies. Calibration history earns a place only where it helps characterize the route relations, the five features, or their stability under perturbation.

A second component test varies checking-route independence (feature 3.). Compare a checker from the generator's lineage with one trained and calibrated separately while approximately matching externally supplied task information, standalone accuracy, interface, and revision policy. This profile predicts greater corrective value from the checker with lower conditional error overlap. An input-coverage account predicts no advantage under the declared boundary; a conditional-coverage account can predict one only by representing the joint error structure that checking-route independence names. Online Resource 1 supplies the full controls and measurement boundary.

The supplementary resource also specifies a temporal test that compares a frozen record with a live route to the same source under update and no-update arms. The design operationalizes live answerability but doesn't distinguish the profile from a dynamic coverage account, which predicts

the same pattern from the information available at follow-up.

Failure of the declared route-specific interactions or update effects, after confirming that the relevant route was invoked and its output reached production in time to affect the answer, would defeat this decomposition. The deployment case changed environmental access without measuring error correlation, so it bears on one feature rather than testing the full decomposition.

Observing that every grounded arrangement in some sample also corrects well wouldn't show that grounding entails corrective control: both might depend on a third architectural feature, or every arrangement sampled might satisfy more than the grounding conditions require. Non-entailment isn't refuted by co-occurrence.

8 CONCLUSION

The octopus and Izzy support opposing verdicts on reference, but this paper's question survives either one. Representations can have content, reference, and correctness conditions while the output-producing arrangement lacks live, sufficiently independent routes from detected discrepancies to altered production. Grounding doesn't entail corrective control.

Language-model arrangements expose the dissociation at scale. Their language models inherit patterns shaped by testimony and coherence constraints. Those patterns supply derivative answerability where source practices have already made target-sensitive corrections, while the arrangements may lack live routes that detect a fresh error and carry it back into production. Identifying which routes constrain an arrangement, how they're related, and for which targets shows whether the missing corrective routes have been supplied or only partly compensated for. Rich internal structure can arise where the target domain is densely encoded in the sequence stream without supplying those live routes elsewhere.

A thin truth predicate can coexist with thick truth-tracking practices. Recurrence of a profile can support bounded expectations without live correction; corrective control adds capacities for detecting and repairing fresh discrepancies. None of that architecture need be built into truth itself.

The older epistemologies keep their subject matter. Each can represent relations among the processes it describes, and reliabilists and social epistemologists routinely do. The projective basis here is the arrangement's current route structure together with whatever calibration history bears on its stability under perturbation, rather than a single process assessed against that background. In ordinary cases the routes operate jointly; language models make their separation visible.

The resulting framework compares arrangements rather than assigning systems to fixed classes. Effective coverage can summarize whether an arrangement had usable information, but it doesn't by itself say which intervention to choose or where a gain should transfer.

Retrieval, multimodal input, tool use, expert review, experimental embedding, and action-guided feedback alter an arrangement's route profile. The question is which route they add, whether that route is live in the task being assessed, and where improvement should transfer. The route-by-task interactions aren't unique to this profile. Its stronger burden is to predict held-out combinations of interventions and tasks from a fixed decomposition better than a prespecified input-coverage model. An arrangement can improve along one dimension and remain fragile along another. Routes and coupling are characterized from causal and architectural facts before performance is checked, and the predicted contrasts can fail.

The lesson extends beyond LLMs. Truth-tracking involves more than linguistic competence, in-

ternal coherence, and statistical sensitivity to past assertion. It depends on a broader, often distributed corrective architecture. What matters is which routes constrain an arrangement, how they're related, which are live, and where their absence should matter.

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SUPPLEMENTARY INFORMATION

Online Resource 1: Proposed empirical designs. Detailed specifications for the crossed clinical design, checker-independence comparison, and temporal test of live answerability.

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